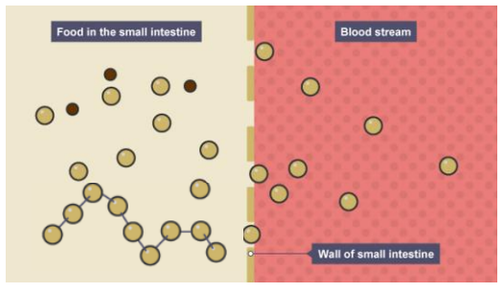
Knowledge Organiser
Year 8: Human Body - Digestion

What is digestion?
- Digestion is the **breaking down** of food we eat into other substances
- Our body cannot absorb or use food without breaking it down
- It can **mechanical** or **chemical**

Diagram which shows the **movement of food molecules** into the blood through digestion

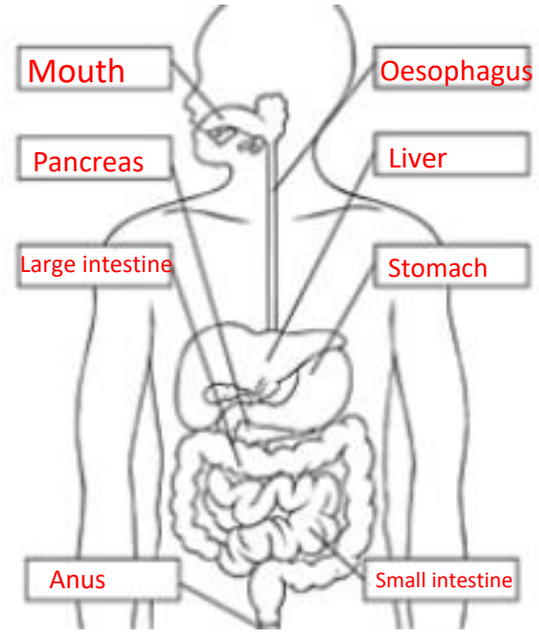


Absorption?
Without breaking them down, food molecules are too big to fit through holes in our small intestine

Mechanical vs. chemical digestion?
- Mechanical digestion involves physical movements
- It includes chewing and muscle contractions
- Chemical digestion uses **enzymes** to break down food

Diffusion into the blood?
- Diffusion is the **movement of particles** from an area of **high to low concentration**
- When we eat, food molecules (e.g. sugars) are at a high concentration in our small intestine
- Our blood contains a low concentration of food molecules
- So food molecules move **by diffusion** across the wall of the small intestine.

Diagram showing the labelled parts of the human **digestive system**



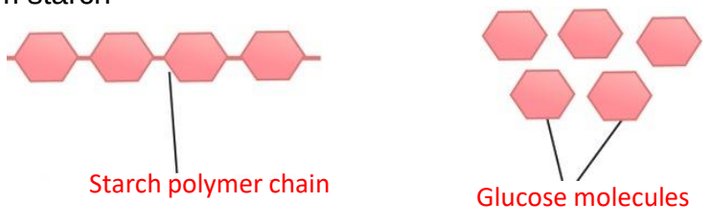
Main biological molecules:
- There are three main molecules we need to extract from the food we eat
- They are carbohydrates, proteins and lipids (fats)
- They are part of larger food molecules which we must break down to absorb and use

	Uses in the body	Found in?	How to test for?	Positive test result?
Starch	Provides energy	Grains, rice, pasta, bread, potatoes	Add iodine	Yellow -> Black/Blue
Lipids	Store of energy, building cell membranes	Cheese, milk, butter, oils	Add ethanol and then water	Colourless → White emulsion
Proteins	Growth and repair of cells	Fish, meat, eggs, beans	Add Biuret reagent	Blue -> Purple

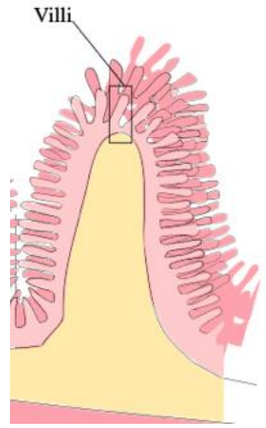
What is starch?
- Starch is a type of complex carbohydrate
- If starch is broken down using an enzyme, the glucouse can be tested for using Benedict's reagent
- It will turn from blue -> brick red if glucouse is present

What are polymers?
- Polymers are long and repeating chains of the same molecule stuck together
- Starch is a long polymer of glucose molecules

Diagram showing the **polymerisation of glucose** to form starch



How is the small intestine adapted for absorption?
- The small intestine is where most digested food is absorbed
- The wall of the small intestine is covered with villi
- These are tiny, finger-shaped structures that **increase the surface area**
- They are only 1 cell thick, allowing for easy **diffusion** into the blood



RAG Yourself!

Learning Outcomes;	RAG
Define diffusion and give examples of diffusion in the human body	
State the main biological molecules and their uses	
Describe how to test for starch and sugars	
Label the digestive organs on a diagram	
Explain the importance of digesting food	
Explain the importance of villi in the intestines	

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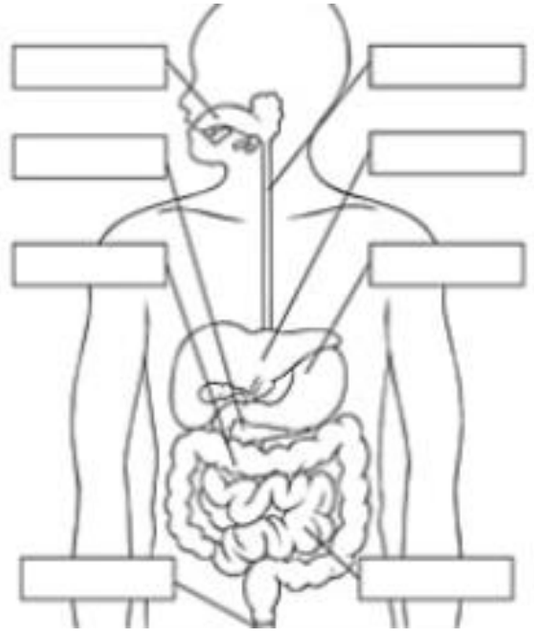
Knowledge Organiser
Year 8: Human Body – Digestion

What is digestion?

Diagram which shows the **movement of food molecules** into the blood through digestion

Absorption?

Diagram showing the labelled parts of the human **digestive system**



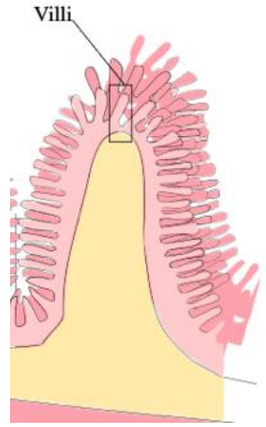
What is starch?

What are polymers?

Diagram showing the **polymerisation of glucose** to form starch

Main biological molecules:

How is the small intestine adapted for absorption?



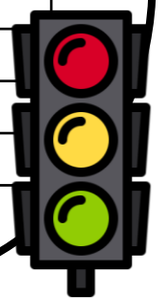
Mechanical vs. chemical digestion?

Diffusion into the blood?

	Uses in the body	Found in?	How to test for?	Positive test result?
Carbohydrates				
Lipids				
Proteins				

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